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Dept: Math

Course: Math467 - Modern Geometry

Assignment: Writing Assignment II

Prompt: This will be between 1000 and 1500 words long (depending on how long your Assignment 1 paper was, so that the total is 2000 words). You should focus on ways in which you personally use geometry in your life. This might be during a job (such as carpentry/construction or graphics design), art, teaching/tutoring, or your general life as a mathematician and ways that connect you personally to geometry. It need not be on a single topic, but should have a theme running through the topics that connects your paper internally (i.e., not a “stream of consciousness”). Pre-approval of topic is not required, but suggested in case you need help focusing your efforts.

Professor: Vince S.

The Geometry of My World: A Personal Connection to Applied Mathematics

As an Applied Mathematics major with a focus on Computational Science, geometry has woven itself seamlessly into the fabric of my daily life and studies. It isn't just a set of abstract principles confined to textbooks; rather, it is an active participant in how I view and interact with the world. From solving practical problems in programming to finding beauty in visualizations, geometry shapes my perspective in ways both pragmatic and poetic.

Geometry, for me, serves as a bridge between the theoretical and the tangible. During my coursework in computational science, I often engage with algorithms that rely heavily on geometric principles. A classic example is the role geometry plays in computer graphics. Tasks such as rendering a three-dimensional object on a two-dimensional screen hinge on an understanding of coordinate transformations, vector spaces, and projection matrices. This interplay between dimensions is not merely a mathematical exercise but a crucial process in making sense of visual data. More generally, we use geometry all the time to convey and break complex ideas down. For example, we often use the set theory principles of intersection and union when developing algorithm conditionals, while being able to visualize these with geometric ideas.

The jump between numerical computers and image based computers relied greatly on the application of geometry to computer systems to make the user experience more engaging. This has given me a personal connection as I see how computers have directly used geometry to make our lives easier. When I worked as a User-Interface designer I had to understand what geometric shapes look most appealing to the human eye, and it turns out that humans want to see more

curves than edges. This led me to creating a company's logo into a more roundish design, and falls along the trend of large companies to favor rounder logos in recent years.

Beyond academic applications, I find joy in geometry's role in creative outlets. As someone who dabbles in graphic design, I often think of shapes, symmetry, and proportions in my work. When designing posters or digital art, the golden ratio—a geometric principle revered for its aesthetic appeal—guides my layouts. Whether it's positioning elements on a canvas or adjusting the proportions of text and images, these geometric considerations transform ordinary designs into visually harmonious compositions. A famous example that comes to mind, is the design of the video game character Sonic The Hedgehog. As his quills sport the iconic golden-ratio curve mentioned prior and continues to be a popular icon 30 years later. More recently, one of my favorite projects was creating a digital illustration of a cityscape. Using isometric drawing techniques, I crafted buildings and streets that appeared three-dimensional on a flat screen. The process required meticulous attention to angles and scaling, as even a slight misalignment could disrupt the illusion of depth. This hands-on experience reinforced my appreciation for geometry as a tool for both precision and artistic expression.

Another area where geometry intersects with my life is in the study of motion and mechanics. As a student of computational science, I frequently encounter problems involving dynamics—how objects move and interact in space. These problems often demand an intuitive understanding of geometric principles such as vectors, angles, and trajectories.

For instance, in a robotics class in 8th Grade, I worked on programming a robotic arm to perform specific tasks. The challenge lay in defining the arm's movement through a series of joint angles and linkages. This required solving inverse kinematics problems, which involve determining the

angles needed to position the arm's end effector at a desired point. The process reminded me of solving a puzzle, where each piece—each geometric calculation—had to fit perfectly to achieve the desired outcome.

Even in everyday life, geometry has proven to be an invaluable ally. From arranging furniture in my apartment to optimizing packing strategies for travel, I rely on spatial reasoning and geometric principles. A particularly memorable instance was when I helped a friend redesign their small kitchen. By sketching a scaled floor plan and experimenting with different layouts, we were able to maximize counter space and improve the workflow.

These experiences underscore how geometry isn't confined to academic pursuits but is a practical tool for problem-solving. It shapes the way I perceive and navigate physical spaces, enabling me to approach challenges with clarity and confidence.

As someone deeply interested in computational science, I often work with data visualizations, another area where geometry plays a pivotal role. Visualizing complex datasets involves transforming raw numbers into intuitive graphs and charts. Whether it's plotting points in a scatterplot or designing a network graph, the underlying geometric principles—coordinate systems, scaling, and symmetry—are crucial to creating clear and impactful visuals.

One particularly challenging project involved visualizing a high-dimensional dataset. Using techniques such as principal component analysis (PCA), I reduced the data's dimensionality and represented it in two-dimensional space. This geometric transformation not only made the data more comprehensible but also highlighted patterns and correlations that were otherwise hidden.

While much of my engagement with geometry is practical, I also find myself marveling at its presence in nature. From the spirals of a sunflower's seeds to the crystalline patterns of snowflakes, geometry reveals itself in countless forms. This natural order inspires me as both a mathematician and a creative thinker.

During a hiking trip, I was struck by the tessellated patterns of basalt columns in a cliff face. These hexagonal formations, created by cooling lava, reminded me of Voronoi diagrams—geometric constructs that partition space based on proximity. Such moments deepen my appreciation for geometry as a universal language, capable of describing phenomena both artificial and organic.

The theme that ties these experiences together is geometry's dual role as a practical tool and a source of inspiration. Whether I am coding a simulation, designing a graphic, or marveling at nature's patterns, geometry enriches my understanding of the world. It provides a framework for solving problems, expressing creativity, and exploring the connections between seemingly disparate disciplines.

As I continue my journey in applied mathematics, I know that geometry will remain a constant companion, guiding my efforts and shaping my perspective. It is more than just a subject; it is a way of thinking—an ever-present reminder of the beauty and order that underpin our complex and dynamic world.